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ORIGINAL ARTICLE

Analyzing indirect economic impacts of wildfire damages on regional economies

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Abstract
This article estimates the economic impacts of wildfire damage on Korea's regional economies, developing an integrated disaster-economic system for Korea. The system is composed of four modules: an interregional computable general equilibrium (ICGE) model for the eastern mountain area (EMA) and the rest of Korea, a Bayesian wildfire model, a transportation demand model, and a tourist expenditure model. The model has a hierarchical structure, with the ICGE model serving as a core module to link to three other modules. In the impact analysis of a wildfire, three external shocks are injected into the ICGE model: (1) the wildfire damaged area derived from the Bayesian wildfire model, (2) changes in travel times among cities and counties derived from the transportation demand model, and (3) variations in visitors' expenditures derived from the tourist expenditure model. The simulation shows that the gross regional product (GRP) of the EMA would decrease by 0.25% to 0.55% without climate change and by 0.51%–1.23% with climate change. This article contributes to the development of quantitative linkages between macro and micro spatial models in a bottom-up system for the impact analysis of disasters, integrating a regional economic model with a place-based disaster model and the demands of tourism and transportation.

KEYWORDS
CGE model, disaster assessment, economic impact, tourism expenditure, wildfire vulnerability

1 | INTRODUCTION

Wildfires have both positive and negative effects on regional economies. The positive effects are generated through demand-side shocks such as increases in fire suppression spending and post-fire recovery and restoration and rebuilding of damaged areas and lost infrastructure facilities, but are expected to be much lower than the negative ones including direct costs of firefighting, loss of regional income from local production, land and property damages, disconnection of infrastructure network, reduction in number of tourist visits, and increases in public health and safety costs for emergency services (Diaz, 2014). For example, the wildfires in the US West in 2018 burned more than 2.3 million ha of the land and a California wildfire has forced road closures in the national forests including parts of Yosemite Valley, which had 4.3 million visitors spending an estimated $452 million in 2017, leads to massive losses in the local tourism revenue (Business Insider, 2018). White ash and smoke spread as far away as 25 miles due to intensifying the fires by a combination of low humidity, high winds, and temperatures. The loss was the worst ever in California, exceeding the loss of 484 residences in the 1961 Bel Air fire. (The City of Portland Oregon, 2010; Los Angeles Times, 2017). The wildfire is expected to become one of the critical disasters in terms of household consumption, production activities, and logistics. Moreover, fire frequency will further increase due to climate change as drier conditions lead to increases in the frequency of extreme events and fire activities (Giglio et al., 2012). The economic damage from wildfires can be classified into direct and indirect damage. The former includes physical losses such as business interruption and unemployment; the latter includes the consequences of industrial and spatial interactions (Cochrane, 2004; Ding et al., 2011; Rose, 2004). However, due to its innate complexities and uncertainties, the indirect damage could not be easily captured without a quantitative method to identify economy-wide effects on markets and economic agents' decisions: this implies that the economic value of damage from disasters has been underestimated due to neglecting research of the indirect impacts on the economy.

This article explores two research questions: what is an indirect economic impact of wildfire damages due to its growing frequency and intensity affected by the climate change? How can we measure such impacts of wildfire damage on the regional economies? Quantitatively assessing the indirect economic damage by natural disasters is core to develope resilient investment and recovery plans to reduce risk. The purpose of this article is to estimate the economic impacts of wildfire damage on Korea's regional economies by developing an integrated disaster-economic system (IDES). This system is composed of four modules: (1) interregional computable general equilibrium (ICGE) model, (2) Bayesian wildfire model, (3) transportation demand model, and (4) tourist expenditure model. The primary model of the IDES is the ICGE model developed for two regions, the Eastern Mountain Area (EMA) and the rest of Korea (ROK), on the base year of 2013, and linked to the other three models. The IDES is applied to the simulation of measuring the loss of regional income due to wildfires in a mountainous area of Korea, the EMA, where large wildfires have frequently occurred (similar to California).

One of academic contributions of this article is to formulate a methodological framework that can examine the indirect economic effect of wildfire damage to take into account of physical flows of raw materials and financial ones of tourism expenditures. Another one is to develop a general equilibrium model to integrate the transportation model with economic model and the Bayesian model on wildfire frequency and intensity under the climate change, deviating from prevalent partial models. In addition, the counterfactual analysis using the IDES as a quantitative tool can provide policy authorities and planner with an economic rationale on how the budget can be allocated into wildfire suppression and recovery sector. The next section focuses on an overview of the application tools for analyzing the economic impacts of disasters on regions. Section 3 estimates the economic impacts of wildfires on regional economies in Korea, developing the IDES. The final section summarizes the findings and offers suggestions for further research.

2 | LITERATURE REVIEWS

2.1 | Economic impact analysis of disasters

Many works have focused on estimating the economic effects of disasters using analytical simulation models (Greenberg et al., 2007; Okuyama, 2007; Okuyama & Santos, 2014; Lee and Moon, 2019; Yeom and Jung, 2019; Yang and Yoon, 2020; Jang et al., 2021; Shin et al., 2021; Hwang and Yoo, 2022). The input-output (IO) model and the computable general equilibrium (CGE) model have been widely used in assessing economic loss from damage in Table 1. The IO model is primarily used in demand analysis and to analyze changes in economic structures arising from disasters. Ryu and Cho (2010) estimated the indirect economic damage due to typhoons and heavy rains in Korea, developing an 'event matrix' designed to calculate a new IO structure after the outbreak of a disaster. Rose et al. (1997) estimated the regional economic losses from earthquake-damaged electric utility lifelines in the New Madrid Seismic Zone in Tennessee using IO and linear programming models. They found that the potential production loss over the recovery period could amount to as much as 7% of the gross regional product (GRP) of the state. They showed how losses could be reduced by reallocating electricity resources and optimizing recovery sequences through linear programming. Okuyama (2004) applied Miyazawa's extended IO framework to estimate the spatial impacts of Japan's Great Hanshin Earthquake of 1995 and the recovery process. He used the sequential inter-industry model (SIM) to investigate the dynamic process of the impact paths of the disaster. Gordon et al. (1998) applied the Southern California Planning Model, an IO-based model, to estimate business interruption costs of the 1994 Northridge earthquake, and found that business interruption accounted for 25%–30% of the total costs of the earthquake. Using a multi-regional input-output (MRIO) model, in den Bäumen et al. (2015) estimated the total indirect loss of production possibilities in the national and global economy due to a 2013 flood in Germany. Most severely indirectly affected by losses in production were the real estate service sector, transport equipment production, and other business services in Bayern. They proposed incorporating the secondary effects of disasters into risk management planning, which would improve a regional economy's prospects of recovering from supply chain restrictions from neighboring economies.

Another type of approach is the integration of the IO model with a transportation network and planning model. Cho et al. (2001) and Sohn et al. (2003) used an integrated model to analyze the economic impact of earthquakes on the transportation network in the Midwest and West Coast US states. The modeling system of Sohn et al. (2003) included a transportation network loss function, a final demand loss function, and an integrated commodity flow model. The article showed that the economic significance was determined not only by the level of disruption but also the volume of flow at the network link, its relative location (topology) on the highway network, and the strength of the economic activities near the network link. Cho et al. (2001) developed an integrated and operational model to trace the effects of an earthquake on the Los Angeles economy and transportation services. It also incorporated bridge and other structure performance models, transportation network models, spatial allocation models, and IO models into a composite operational system. They found that the spatial distribution of these losses was sensitive to changes in network costs caused by transportation structure losses. Kim et al. (2002) estimated the impact of an earthquake on transportation cost using IO analysis integrated with a transportation demand model based on the US Interstate Highway system. The regional impacts of transportation disruptions caused by the earthquake were calculated in three scenarios; one of the three evaluated the reduction in overall shipment cost as 186 million ton-mile/year and an average of 10.82 miles. The results provided a basis for making policy decisions to mitigate unexpected disasters and for planning to construct new highway network sections to strengthen the existing network. Transportation network and the National Interstate economic Model (TransNIEMO) is a combined model of the IO model with a freight network to support the analysis of the economic consequences of an attack on

TABLE 1 Impact analysis of disasters

Author	Type of disasters	Model	Impacts/Key issues
Rose et al. (1997)	Earthquake	IO model and Linear programming	Reduction in the GRP by 7%
Cho et al. (2001)	Earthquake	IO model	Integration of network model, spatial allocation model and the IO model
Kim et al. (2002)	Earthquake	IO model	The IO model combined transportation model
Sohn et al. (2003)	Earthquake	IO model	Network effects on transportation
Okuyama (2004)	Earthquake	Sequential inter-industry model	Impacts on inter-regional and inter-industrial sectors
Rose et al. (2005)	Disruption in water service	CGE model	Impacts of water service disruptions
Bosello et al. (2007)	Sea level rise	CGE model	Impacts on the GDP and energy consumptions
Tatano and Tsuchiya (2008)	Earthquake	Spatial CGE model	Direct and indirect spillover effect on regional economies
Ryu and Cho (2010)	Typhoon and heavy Rain	IO model	Reduction in the GDP by 1.18%
in den Bäumen et al. (2015)	Flood	MRIO model	Indirect loss of production €6.2 billion
Baghersad and Zobel (2015)	Disasters	New linear programming model with IO system	Indirect economic impacts of disasters
Koks and Thissen (2016)	Floods	IO model	Supply driven regional IO model with transport disruption
Husby and Koks (2017)	Disasters	The IO and the CGE model with ABMs	Integration of micro model with the IO and the CGE model

Source: revised from Kim and Kwon (2016).

the bridge and truck services of the highway in the United States (Park et al., 2011). The integrated model with micro models based on the IO model has been developed such as NIEMO series, and the main subjects of study were economic effects analysis caused by natural and manmade disasters such as hurricanes, terrorism, and transportation network disruption(Cho et al., 2015; Park et al., 2016; Richardson et al., 2005). Koks and Thissen (2016) estimated the economic impacts of three floods of Rotterdam in the Netherlands, using an integrated recursive dynamic multiregional supply-use model with production technologies and constraints on the supply of transportation services. Baghersad and Zobel (2015) developed a linear programming model based on the IO model to assess the economic consequences of production capacity bottlenecks caused by disruption of the electricity sector in Singapore. In 12 different scenarios, the total average inoperability and total inoperability of the entire economy would diminish, with decreasing initial loss and increasing recovery in the electricity sector. This finding implied that improving the robustness of the electricity sector could have a stronger impact on the total average inoperability of the economy than decreasing the recovery time.

The CGE model has succeeded in transforming a complex and abstract economic structure into a numerical and analytic framework of supply-and-demand-side channels based on microeconomic optimization of economic agents using a variety of assumptions and macroeconomic-closure rules. Bosello et al. (2007) used the CGE model to measure the amount of land and capital loss due to sea-level rise in coastal regions. Gross domestic product and energy consumption would decrease in the absence of coastal protection, but certain regions with substantial dike building could increase their GRP level despite a reduction in the utility level with coastal protection. Tatano and Tsuchiya (2008) presented a spatial CGE model to estimate the economic losses incurred due to transportation network disruption after Japan's Niigata-Chuetsu earthquake. The simulation showed spillover patterns of economic losses arising from the earthquake concerning intra and interregional trade. This paper discussed counter-measures to reduce negative spillovers, such as adopting mitigation policies for the reduction of damage to residential facilities. Rose and Liao (2005) analyzed the economic impact of the disruption of water services on the Portland (Oregon) Metro economy, focusing on the indirect economic losses resulting from the water shortage, the extent of preevent mitigation, and postevent inherent and adaptive resilience.

While previous studies have integrated economic approaches to estimate physical damage, more work is required on the interactions of economic agents with respect to spatial mobility of factor inputs and commodities using a spatial network. In particular, as discussed in Husby and Koks (2017), the integration of micro-spatial analysis at the city level with traditional macro-spatial or aggregated regional models might be suitable for economic analysis of disasters. It is essential to understand the spatial diffusion patterns of the adverse effects from disasters across spatial units.

2.2 | Economic impacts of wildfire

Cost approaches and multivariate analyses have been widely applied to the economic analysis of wildfires (Rahn, 2009). Mercer et al. (2000) examined the economic impact of catastrophic wildfires and the efficacy of fuel reduction in Florida using spatial and econometric models. The loss from the 1998 Florida wildfires was estimated to be US$622–880 million, including the costs to timberland owners (US$345–605 million), the tourism industry (US$138 million), and resources diverted to fighting the fires (US$100 million). Kunji et al. (2002) assessed the effect of wildfires on air quality and health in Indonesia. More than 90% of respondents had respiratory symptoms, and elderly individuals suffered a severe deterioration of overall health. Between September and November 1997 in Indonesia, there were 527 hazed-related deaths, 0.30 million cases of asthma, 58,095 cases of bronchitis, and 1.45 million cases of acute respiratory infection. Sage and Nickerson (2017) estimated the direct impacts of the wildfire damage in Montana in 2017 on the economy using cost analysis. The state lost up to 800,000 visitors due to the summer's fires

and smoke in 2016, resulting in a loss of US$240.5 million in visitor spending. Pyke et al. (2016) measured the direct impact of the 2013 bushfire in North East Victoria, Australia, due to both the fire and postfire flooding. During the 55 days of burning in early 2013, visitor spending losses were estimated to be approximately US$1.5 million, based on a visitor survey. To minimize the economic effects of fire events, Pyke et al. (2016) recommended that tourism planning and stakeholder communication should be a key priority.

In terms of multivariate analysis, Rahn (2009) estimated the overall economic impact of wildfires on regional economies in San Diego as US$2.45 billion or US$6,500 per acre. The total suppression cost was less than 2% of the entire economic impact, which was a relatively negligible cost in contrast to the overall loss. Moseley et al. (2012) analyzed the effects of large fires in California during 2004–2008 on labor markets. Local employment and wages in one county increased during massive wildfires; the economic impact of the suppression effort outweighed the economic disruption from wildfires in the short term. However, in the long term, the extensive wildfires caused instability in local labor markets by amplifying seasonal variation in the tourism and natural resource sectors. Using the difference-in-difference approach, Kiel and Matheson (2018) studied the Fourmile-Lefthand Canyon forest fire in Colorado 2010. They showed that the sale price of housing could decline by as much as 21.9% due to such fires. Kochi et al. (2016) measured the economic cost of wildfire smoke exposure in terms of the medical cost of hospital admissions in 2007 using a time-series count model and a negative binomial model: the total medical cost was estimated at over US$3.4 million.

Previous works have attempted to measure the economic costs of wildfires focusing mainly on forest timber loss, the changes in wildfire suppression budgets, and recovery costs in Table 2. However, most articles did not take into account the secondary or indirect damage to the regional economies, such as the decrease in production activities, the increase in travel times, and the losses to tangible and intangible cultural heritage such as tourism resources (Marrion, 2016). It is necessary to measure not only the indirect effects of disasters on a region and its neighboring areas but also the harmful effects of its subsequent damage, including flooding and smoke, on property loss and human health in terms of sectoral linkages. Besides, previous articles on wildfires have identified changes in the forest appearance and the damage to forest trees and crops. Since the damage caused by the wildfires depends on geographical and environmental factors, spatial microanalysis is the key to the analysis of the probability of the occurrence of wildfires and the spatial diffusion area, developing disaster prevention programs for an early-warning system for wildfires and thus decreasing the amount of economic damage.

3 | ANALYSIS

3.1 | Model

As discussed in the previous section, the spread of wildfires tends to decrease spatial accessibility levels and, consequently, economic activities of service-oriented industries such as accommodation, food services, wholesale and retail trades, transportation, and storage. It implies that the economic analysis of widlfires requires implemention of an economy-wide or multisectoral model to be integrated with a few key modules to calibrate occurrences of wildfires and changes in transportation demands and to estimate indirect effects of such natural disasters on production and consumption activity by the economic agent. In this article, IDES is developed for measuring indirect economic effects of wildfire damages on regional economies, and is composed of four modules: (1) the ICGE model for economic analysis of macro-regions, (2) the transportation demand model for calibration of travel times, (3) the tourist expenditure model for estimating visitors' spending, and (4) the Bayesian wildfire model for estimating the damaged areas, as shown in Figure 1. This integrated approach for the economic analysis of wildfires is the novelty of this article in a sense that traditional methods used in this subject have been rooted in partial equilibrium and linear-demand side models as summarized in Chapter 2. In terms of computational process, the first step is to measure annual wildfire damage size using the disaster model (i.e., the Bayesian wildfire model) with climate and topographical factors. The next step is to calculate the travel time increases and accessibility decreases in response to the occurrence of wildfires using the transportation demand model. The third step is to calibrate the reduction in visitors' expenditures using the tourist expenditure model. The final step is to estimate the negative effect of the wildfires on the regional incomes, injecting (1) the damaged area of the disaster model, (2) the spatial accessibility level of the transportation demand model into the production function, and (3) the changes in the visitors' expenditures of the tourist expenditure model into the household consumption in the ICGE model.

TABLE 2 Impact analysis of wildfires

Author	Country/Region	Method	Impacts
Mercer et al. (2000)	USA/Florida	Spatial and econometric analysis	US$1864 per acre of economic losses
Kunji et al. (2002)	Indonesia	Econometric analysis	Change in mortality
Rahn (2009)	USA/California	Cost analysis	US$2.45 billion of costs
Moseley et al. (2012)	USA/ California	Econometric analysis	Increases in local employment and wages
Kiel and Matheson (2018)	USA/Colorado	Econometric analysis	Decline of housing sale price by 21.9%
Kochi et al. (2016)	USA/California	Econometric analysis	US$3.4 million of medical costs
Pyke et al. (2016)	Australia	Cost analysis	US$1.5 million of postfire flooding costs
Sage and Nickerson (2017)	USA/Montana	Cost analysis	US$240.5 million of visitor spending losses

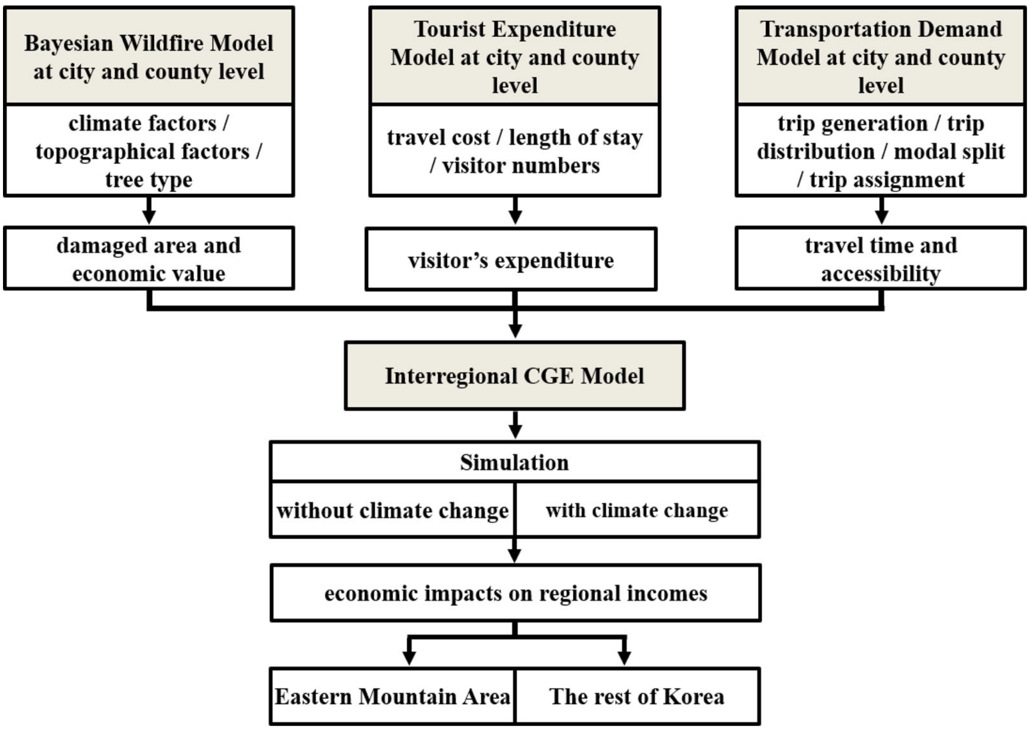

FIGURE 1 Structure of integrated disaster-economic system.

The core of the entire system, the IDES, is the ICGE model used to account for maximization of a producer's profit and a consumer's utility in the real side economy, following market-clearing prices under equilibrium, as shown in Table 3. In the model, there are two macro-regions, EMA and the Rest of Korea (ROK), as well as one representing the rest of the world (ROW). Production activity is divided into 12 sectors: four forest sectors (forest products, wood and wood products, pulp and paper products, and processing of timber), five tourism sectors (retail and wholesale, transportation, restaurants and accommodation, cultural services, and sports and entertainment services), primary, manufacturing, and service sector. Each industrial sector is assumed to produce a single commodity, which is disaggregated into an intraregional supply, a regional export, and a foreign export by production destination. There are three types of commodities: an intraregional supply (demand), a regional import, and a foreign import in the regional market in terms of the product origin. Each commodity and factor input price is determined through an equilibrium process between supply and demand in each market. The basic structures of the CGE model have been discussed in many textbooks including Shoven and Whalley (1992), so rather than repeating such details here, we focus on the points that distinguish our model from previous ones.

The production structure model is composed of three-stages. At the top of the structure, the gross output is determined via a Leontief production function of value-added and intermediate inputs. The latter is derived from interregional IO coefficients, whereas the value-added at the province level is simply the sum of those of cities and counties. These are specified with a spatial econometric Cobb-Douglas production function of labor and capital inputs with the land area variable as a proxy for development

TABLE 3 Major equations of interregional computable general equilibirium (CGE) model

Output	Output = Leontief (Value added, Intermediate demand)
Value added	Value added = Total Factor Productivity*CD (Capital stock, Labor, Land)
Supply	Output = CET (Foreign exports, Domestic supply)
Domestic supply	Domestic supply = CET (Regional exports, Intraregional supply)
Demand	Demand = Armington (Foreign imports, Domestic demand)
Labor demand	Labor demand = LD (Wage, Value added, Net price)
Total factor productivity	Total Factor Productivity = TFP (Accessibility, Population)
Labor supply	Labor supply = LS (Labor market participation rate, Population)
Population	Population = Natural growth of population + Net population inflows
Regional incomes	Regional incomes = Wage + Capital returns + Government subsidies
Migration	Migration = TODARO (Incomes and employment opportunities of origin and destination, Distance between origin and destination)
Consumption	Consumption by commodity = CC (Price, Incomes)
Private savings	Private savings = PS (Saving rate, Income)
Government revenues	Government revenues = Indirect tax + Direct tax + Tariff
Government expenditures	Government expenditures = Government current expenditure + Government savings + Government investment expenditure + Government subsidies
Labor market equilibrium	Labor demand = Labor supply
Capital market equilibrium	Private savings = Total investments
Commodity market Equilibrium	Supply of commodities = Demand for commodities
Government	Government expenditures = Government revenues

Source: revised from Kim and Kwon (2016).

potential. In the second stage, the intermediate demands are transformed into demands for domestic products and imports using the Armington approach. The total demands for the domestic products

are classified into intraregional supplies and regional exports, which are determined by their relative prices and spatial interaction between two regions in the third stage. Also, the profit maximization under the two-level constant elasticity of transformation (CET) function produces an optimal allocation of the gross output via the domestic supplies and foreign exports. Each regional labor input is assumed to be homogeneous and moves among the regions, while capital stock cannot move from one region to another in the short run. The labor supply relies on the participation rate and the total population size of the region overall. The population is the sum of the natural increase of the population combined with the net gain (or loss) of the migrant population. The latter is assumed to respond to interregional differences between origin and destination regions in terms of wage per capita and unemployment rate, as well as the physical distance between the regions.

The total demand consists of intermediate and final demands such as consumption and investment expenditures of private and government sectors. Two tiers of government structure are specified in the model: two macroregional governments and one national government. Government expenditures are composed of the consumption expenditures, subsidies to producers and households, and savings. Revenue sources include taxation of household incomes, value-added, and imports. Aggregate savings are the sum of household savings, corporate savings of regional production sectors, private borrowings from abroad, and government savings while determining total investments. There is one consolidated capital market without financial assets in the model, and the numeraire of the model is set as the consumer price index. In addition, a social accounting matrix (SAM) is calibrated as a benchmark for the development of the interregional CGE model. The SAM consists of six accounts—factors, households, production activities, government, capital, and the rest of the world—and is treated as an initial equilibrium for the Interregional CGE model. As shown in the structure of the ICGE model in Figure 2, the regional economic effects heavily depend on the regional and sectoral linkages and the type of macroeconomic closure rules. It is not easy to assess the validation and prediction performance of the CGE model due to its deterministic foundation: in general, the model's reliability has been evaluated in terms of measuring the stability of results over time (De Maio et al., 1999; Kim et al., 2016). In this article, a sensitivity analysis of various model settings and critical parameter values was performed to determine the robustness of the results as discussed in Belgodere and Vellutini (2011) and Kim et al. (2016). The GDP and the GRP of the EMA could be reduced by 0.1%–0.3% and 0.2%–0.5%, respectively, if the elasticities of substitution and transformation for the Armington and CET functions are increased by 10%: this means

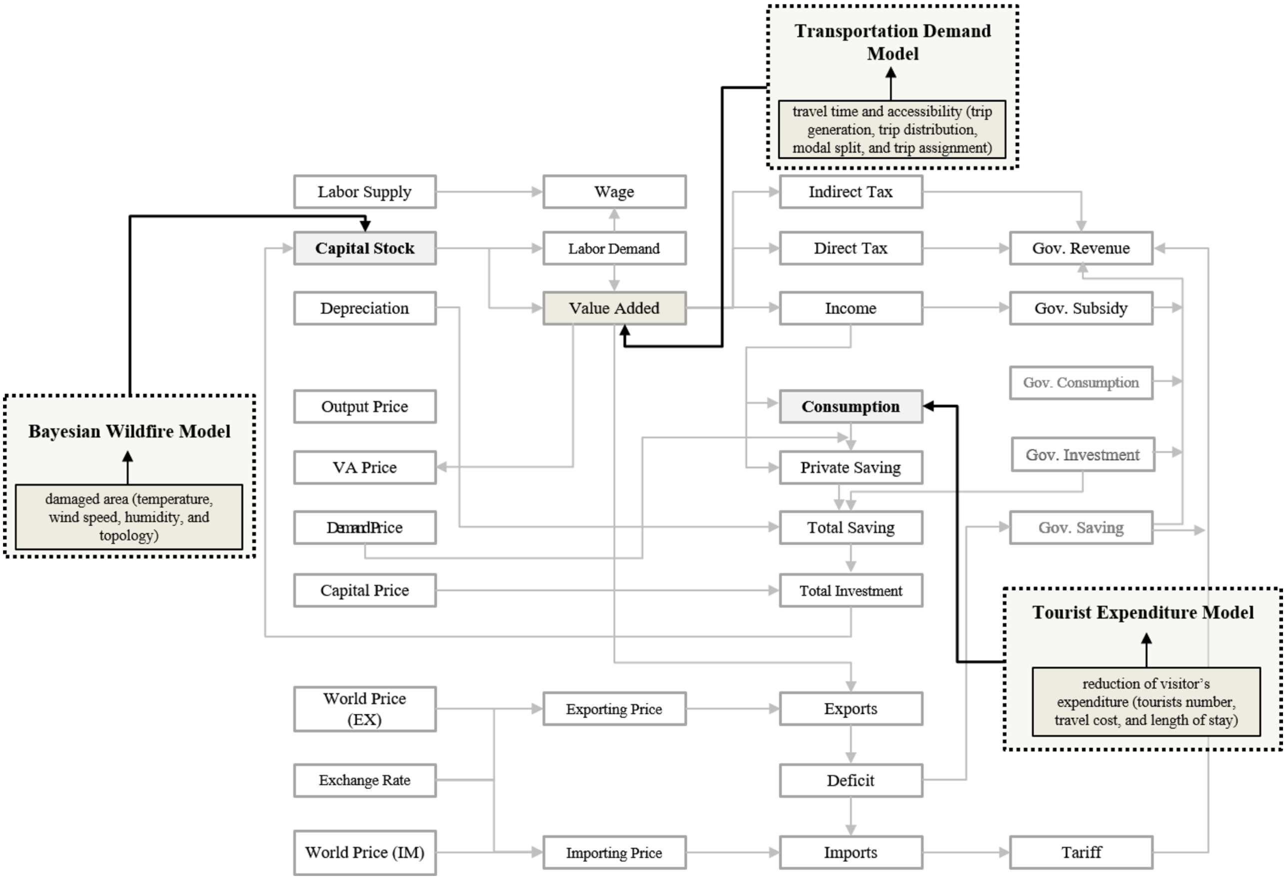

FIGURE 2 Structure of interregional computable general equilibrium model.

that the model is comparatively reliable for counterfactual analysis.

The second module of the IDES is the wildfire model to calibrate the wildfire damaged area using climate and topographical factors. The model in this article is estimated with the Bayesian estimation method: Mori and Johnson (2013) discussed the suitability of the Bayesian model for simulating the risk of wildfires. As a robust method, the Bayesian approach can estimate a posterior probability for the extent of wildfire damage assuming prior probability and infers the uncertainty or variability of parameters derived from the autocorrelated or nonrandom data (McCarthy, 2007; Mori & Johnson, 2013; Jaafari et al., 2017). The independent variables in the disaster model are maximum temperature, humidity, wind speed, the slope of topography, and pine tree ratios as independent variables; human factors cannot be included in the model due to data limitations (Woo, 2015). Meteorological observations (maximum temperature, relative humidity, and average wind speed) and topographic features (slope and pine tree ratio) are provided by the Korea Meteorological Administration and the Korea Forest Service, respectively. The spatial analysis tool, ArcGIS, is used to examine the slope and pine tree ratio by grid zone. The contour layer is extracted using the triangulated irregular network (TIN) on the digital topographic map of the forest, and the inclination is calculated by constructing a zone from the digital elevation model (DEM) to the grid. The calculation method for the slope is the average maximum technique, which combines the neighborhood algorithm with the maximum downhill slope algorithm provided by ArcGIS10.1. The parameters of wildfire model are estimated by creating 2,417 lattice vectors (highest temperature, average wind speed, relative humidity, and slope) in the EMA, as shown in Equation 1.

$$DA_i = 2.060 + 0.332\,HT_i + 1.888\,WS_i - 0.147\,HU_i + 1.250\,PT_i + 1.938\,CFS_i \quad (1)$$

$$(22.637^{***}) \quad (4.049^{***}) \quad (16.561^{***}) \quad (-4.083^{***}) \quad (11.905^{***}) \quad (15.260^{***})$$

$$R^2 = 0.710$$

$$({*}\ p < 0.10.\ {**}\ p < 0.05.\ {***}\ p < 0.01)$$

where DA_i is the wildfire damaged area (ha), HT_i is the highest temperature (°C), WS_i is the average wind speed (m/s), HU_i is the relative humidity (%), PT_i is the ratio of the pine tree in the area (%), CFS_i is Cos converted facing slope (degree).

The next module is the transportation demand model to generate the shortest travel time (minimum travel distance) between each city and county through a mathematical process of trip generation and distribution, and modal split with the assignment. The shortest route algorithm in the network assignment results in a set of the shortest travel time, travel speeds, and travel demands on the links in the network using the EMME 4 program. The user balance principle (Wardrop, 1952), which yields a balance of demand and supply, is applied to the transit assignment, while an optimization algorithm of the link impedance (cost) function is used to derive the equilibrium state. The shortest travel time from the transportation demand model can be an input value for accessibility, which is defined as the spatial interaction or development potential contacts with the levels of nonwork trip and work-trip activities at the origin(O) and destination (D) as a gravity type (Kim et al., 2004). This accessibility level is injected into the production function of five industries and tourism consumptions at the macro region scale in the ICGE model.

The last module in the IDES is the tourist expenditure model to estimate spending using the multivariate regression framework in terms of supply and demand factors (Song & Li, 2008). The primary independent variables of tourist expenditures in the EMA are a visitor's income, daily travel cost for the previous year, the number of firms at a destination, and road accessibility. The equation is estimated based on the Korea Domestic Tourist Survey and transportation demand information for 2013. The road accessibility is a weighted average of the population of the node and link impedance (travel time distance) between the 236 origin and 18 destination regions in the EMA as a proxy for transportation services in the tourist expenditure model (Kim et al., 2004, 2017). The occurrence of a wildfire could decrease road accessibility for a trip to the EMA, which would reduce total tourist expenditure at destinations in the EMA. For instance, if the accessibility drops by 1%, the expenditure would reduce by 0.030%, as shown in Equation 2.

$$\begin{aligned}&?lnTE?^t_i = -1.159 + 0.222?lnINC?^t_i + 0.055?AGE?^t_i\\&-0.001??(AGE?^t_i)?^2 + 1.392ln?TC?^{(t-1)}_i\\&+0.203ln?EMP?^t_D(? - 0.670?^{*})(?3.450?^{(***)})(?5.110?^{(**)})\\&(? - 5.980?^{(***)})(?28.720?^{(***)})(?2.570?^{(***)})\\&+0.111?lnHTL?^t_D + 0.316ln?FIRM?^t_D + 0.170ln?CULT?^t_D\\&+0.078ln?POP?^t_O + 0.030ln?ACC?_OD^t + ?_i\end{aligned} \tag{2}$$

$$\left(2.450^{**}\right)\left(2.970^{***}\right)\left(1.990^{**}\right)\left(2.200^{**}\right)\left(1.820^{*}\right)$$

$$R^2 = 0.645$$

$$(*\ p < 0.10.\ **\ p < 0.05.\ ***\ p < 0.01.)$$

where TE^t_i is the expenditure of individual tourist, INC^t_i is the personal income of individual tourist, AGE^t_i is the age of individual tourist, TC^{t-1}_i is the tourist's daily travel cost at the previous year, EMP^t_D is the number of employees in travel firms in the destination region (D), HTL^t_D is the number of the hotel room in the destination region (D), $FIRM^t_D$ is the number of firms in the destination region (D), $CULT^t_D$ is the number of cultural legacy in the destination region (D),

POP^t_O is the population size of origin region (O), and ACC^t_{OD} is the road accessibility between origin region (O) and destination region (D).

3.2 | Simulation

The IDES is applied to estimate the economic effects of a wildfire on regional incomes. As a case, we assume that the wildfire takes place in Goseong County in the EMA, where more than half of Korea's massive wildfires occur, as shown in Figure 3. This area was the site of the 2018 Pyeong Chang Winter Olympics: it is well known for tourist attractions and is covered with mountainous terrain[1]—as much as 80% of the total area, but has quite low accessibility due to high-sloped mountains above 1,000 m. The economic impacts of a baseline without the wildfire occurrence are compared with those of an experiment with the disaster. The occurrence is sensitive to the wind speed and the humidity of Equation (1) which climate change has substantial influences on, so the experiment is disaggregated into two cases, Experiment 1 without the climate change and Experiment 2 with the climate change. The level of the climate change can be quantified with a representative concentration pathway (RCP) of the fifth assessment report (AR5) of the Intergovernmental Panel on Climate Change (IPCC). IPCC (2014) specified four types of climate futures in terms of a range of climate forcing levels in 2100; RCP2.6, RCP4.5, RCP6, and RCP8.5 (2.6, 4.5, 6.0, and 8.5 W/m^2, respectively). In this paper, Experiment 2 assumes the climate change with RCP8.5, the high emission scenario of greenhouse gas (GHG) (Riahi et al., 2011).

Baseline: No wildfires in the Eastern Mountain Area (EMA)

Experiment 1: Wildfires in the EMA without climate change

Experiment 2: Wildfires in the EMA with climate change under the high GHG emission (RCP8.5)

The IDE model is used to measure how climate change affects the economic losses resulting from wildfires as a counterfactual analysis. In the simulations, three external shocks are added into the IDES: (1) the wildfire damaged area in Goseong derived from the Bayesian wildfire model, (2) changes in travel times among cities and counties due to the wildfire from the transportation demand model, and (3) decreases in visitors' expenditures derived from the tourist expenditure model. That is, the shock to

[1] There were 33 large scale wildfires since the year 2000.

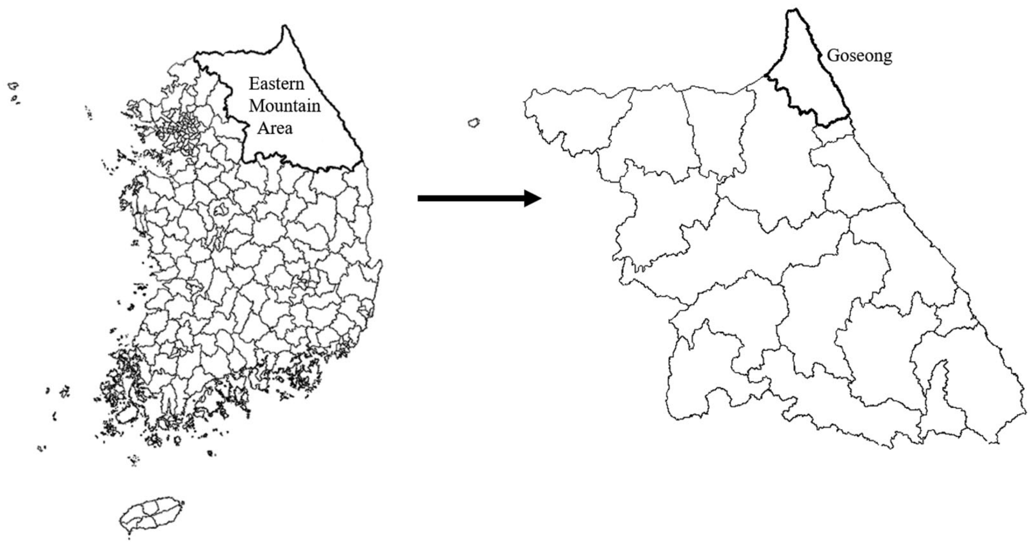

FIGURE 3 Damaged areas of wildfires.

the regional economy in terms of supply-side is a reduction in potential outputs by labor and capital shortages and a malfunction of production systems such as the disruption of production machinery. This damage to the transportation network increases transportation costs or distorts the commodity flows. This spillover effect in turn affects production chains of neighboring regions, which decreases the regional attractions and changes consumer patterns.

These shocks caused by wildfires are expected to result in (1) increases in the damaged areas, (2) reductions in wood inputs, and land stock in the production process of forest and tourism industries, and (3) increases in travel time (distances) by a temporary disruption of road networks. All roads within a radius of 5 km from the damaged area are assumed to be closed for 6 months in the occurrence of a wildfire[2]. Any mobility of vehicles and people, including economic activities, is not allowed in the simulation. The baseline accounts for what would happen without any significant change, and the differences in the damaged area, accessibility, and tourist expenditure between the baseline and the scenario are considered a "shock" to the model. The shock is injected into the ICGE model, after which a new set of equilibrium values can be generated subject to the price normalization rule and the exogenous consumer price inflation rate. Each baseline and counterfactual simulation produces a sequential path of economic behavior following an internal consistency. The impacts are measured in terms of changes in the value-added and the production of goods and services. The results from the counterfactual experiment are then compared with those from the baseline case.

[2] Generally, at least 6 months have been required for recovering economic activities of the damaged areas from the wildfires. For example, transportation flows have been partially restricted due to the 2020 Goseong wildfire for six months. Please refer to the link below. https://www.edaily.co.kr/news/read?newsId=03444006622457432&mediaCodeNo=257&OutLnkChk=Y

The first step for the simulation procedure is to measure the damaged areas in Goseong, using the disaster model. The annual or monthly values of three independent weather-relevant variables (temperature, wind speed, and humidity) have been widely volatile, so each variable is assumed to follow the Gumbel probability distribution based on extreme value theory of Lee et al. (2006) and de Zea Bermudez et al. (2009). The lower and the upper levels of input values of three key variables are determined by Markov Chain Monte Carlo (MCMC) approach as a stochastic simulation model to reproduce a virtual situation (Binder & Heermann, 2010). Therefore, the sizes of damaged areas depend on three combinations of independent variables: each variable has lower and upper limits, so there are eight sets of value choices for the three variables summarized Table 4. It shows ranges of the wildfire damaged areas for two case (with and without climate change) from the results of the Monte Carlo simulations of a combination of lower and upper levels of three stochastic variables (temperature, average wind speed, and relative humidity distributions) in the Goseong area. The average damage area under the climate change of RCP8.5 scenario could increase by 40.4%, compared to the scenario without climate change.

The next step is to quantify the effects of the wildfire on the travel times using the transportation demand model. The approach finds that a wildfire could increase the average travel time of the EMA by 2.42% due to the imposition of a "no passing" zone in the area within a 5 km radius of Goseong. The third step is to estimate changes in consumption expenditures of visitors to Goseong using the tourist expenditure model: the expenditures are determined by multiplying the number of tourists, travel costs, and length of stay. The consumption amounts of the EMA and the ROK residents of Goseong would decrease by 5.78% and by 6.07%, respectively. If a wildfire occurs in Goseong County, there

TABLE 4 Wildfire damaged area by weather condition and climate change

Scenario cases	Without climate change		With climate change (RCP8.5)		Weather condition		
	Lower limit	Upper limit	Lower limit	Upper limit	Temperature	Wind speed	Relative humidity
1	55.3	62.7	80.0	87.0	Lower	Upper	Lower
2	53.7	60.9	74.9	81.7	Lower	Lower	Lower
3	51.9	59.0	69.8	76.5	Lower	Upper	Upper
4	57.8	65.3	82.0	88.7	Lower	Lower	Upper
5	53.7	60.9	74.9	81.7	Upper	Upper	Lower
6	52.3	56.5	69.0	73.8	Upper	Lower	Lower
7	49.2	54.0	66.9	72.1	Upper	Upper	Upper
8	45.8	51.4	64.8	70.4	Upper	Lower	Upper

TABLE 5 Impacts of wildfire on value-added and gross regional product (GRP) of eastern mountain area (EMA) (unit: %)

	Without climate change			With climate change (RCP8.5)		
	Lower	Mean	Upper	Lower	Mean	Upper
Four forest sectors*	−12.12	−14.75	−17.43	−17.43	−20.02	−23.11
Five tourism sectors**	−0.77	−0.74	−0.85	−0.85	−0.99	−1.39
Primary sector	−0.02	−0.05	−0.12	−0.15	−0.18	−0.62
Manufacturing sector	0.02	0.08	−0.11	−0.17	−0.21	−0.75
Service sector	0.19	0.05	−0.09	−0.01	−0.42	−0.70
GRP of EMA	−0.25	−0.37	−0.55	−0.51	−0.84	−1.23
GRP of ROK	0.00	0.00	0.01	0.00	0.01	−0.01
GDP (Total GRP)	0.00	−0.01	−0.01	−0.01	0.00	−0.04

*Four forest sectors: forest products, wood & wood products, pulp & paper products, and processing of timber.
**Five tourism sectors: retail & wholesale, transportation, restaurants & accommodation, cultural services, and sports & entertainment services.

would be a sharp downturn in the GRP as expected in Table 5. The GRP of EMA would decline by 0.25% to 0.55% without climate change and by 0.51% to 1.23% with climate change. This result implies that climate change could magnify the economic loss from 0.26% points to 0.68% points for the EMA, depending on the weather conditions. In addition, a wildfire would reduce value-added in the forest sector and the tourism sector for the EMA: the value added of the forest and the tourism sectors could decrease by 12.12%–17.43% and by 0.77%–0.85%, respectively. Meanwhile, the economic damage at the national level would not be as substantial as possible because the factor inputs and regional trades would continue to move across the sectors and regions, maintaining the stability of economic conditions.

However, the interesting point is that the GRP of ROK could increase by 0.01% for the case without climate change, while the GDP could decrease slightly by 0.01% without climate change and by 0.01%–0.04% with climate change. That is, the ROK would be able to enjoy unexpected benefits by increasing its share in the domestic market in the short run, though the EMA would experience a value-added loss from the disaster. The economic benefits to the region should not be regarded as a positive effect of a wildfire because the increase in the GRP is generated through the sacrifice of the damaged region. This finding shows that it would be worthwhile not only to implement regional coordination and resilience programs to share the costs and benefits from the disaster. For instance, local governments can draw up dark tourism projects for the damaged areas as a source of compensation for economic losses and as a means of promoting wildfire prevention policies. Moreover, they need to launch forming a spatial partnership by jointly investing in the development of unmanned aerial vehicles in order to improve wildfire management system concerning wildfires detection and suppression, and cadastral surveying on large-scale zones (Tang & Shao, 2015). Finally, in a sense that the tourism industry tend to be vulnerable to the natural disaster, it is necessary to focus on industrial diversification of regional economies as cost-effective and risk-mitigating actions.

4 | CONCLUSION AND RESEARCH AGENDA

This article estimates wider economic impacts of wildfire damage on regional economies, developing the IDES for Korea. The key module of this system is the ICGE model which is linked with the other three modules of (1) the Bayesian wildfire model as a place-based disaster model, (2) the transportation demand model, and (3) the tourist expenditure model in terms of the model's hierarchical structure. The application of the IDES to the wildfire case in Goseong County shows that the GRP of the EMA would decrease by 0.25%–0.55% without climate change and by 0.51%–1.23% with climate change. In particular, this analytical tool provides economic agents and policymakers how many financial resources should be allocated to precautionary and rehabilitation activities and forest preservation, taking into account the economic value of forest assets as well as the costs of damage to natural resources, and suppression and recovery costs. Though the IDES has strong assumptions such as market-clearing pricing and optimizing behavior of economic agents, it can be used for implementing an early warning system (EWS) for the disaster planning and prevention programs. The EWS for the wildfire can consist of three components incluidng (1) detection of the wildfire occurrence, (2) prediction of the direction of spatiotemporal spread, and (3) calculation of spatial and economic damages in a systematic information system (Jaya & Folmer, 2021). The ICGE model evaluates economic losses of the third component as discussed earlier, and the the wildfire model to be incorpotated with Digital-Twin (DT) system can trace out the spatial diffusion pattern in the second component. The DT is an intelligent technology for multi-stage management to solve the wildfire prevention, the mitigation and the recovery (Dianyou and Zheng, 2022). The connection of the Internet of Things (IoT) sensors of the DT on the mountain areas and its interactive wildfire spread simulation improve the accuracy of the wildfire model and the fire occurrence probability model with spatial covariates and autocorrelation using a spatial point processing method (Kim and Woo, 2019; Ha et al., 2021; Park, 2021; Shin et al., 2021; Kim and Kang, 2022; Sim et al., 2022).

A few points need discussion regarding the prospect of further research on application and modeling issues. First of all, the model does not capture long-run effects from a sequential process of recovery of production capacity losses on regional economies. In order to measure dynamic effects of the disaster events on industrial and spatial allocations of factor inputs and production activities, it is necessary to modify the current model's structure into backward-looking (adaptive) or forward-looking (perfect foresight) approach: the former produces recursively dynamic outcomes based on a stock-flow of the capital market under a fixed savings rule, and the latter finds optimization behavior for consumers and producers over all the periods in spite of difficulties in solving an optimal set of prices and quantities. In addition, the spatial diffusion pattern of wildfires needs to be examined at a micro spatial unit because weather conditions usually depend not only on natural resources but also the built-environment and structures. For example, in terms of the spatial unit of the IDES, it would be desirable to replace hierarchical system (the provincial level for the ICGE model and the city and county one for the other three models) with a coordinated grid-based system (e.g., equal-size grid cell with 1 km^2). This system contributes to reducing distorted influences of administrative boundaries such as border effects and in identifying spatial interactions among environment and economic agents using GIS (EUROSTAT, 2021). The final topic is to calibrate the wildfire damaged areas with a stress test, as a special case of the simulation, on the climate change and the wildfire occurrence. In this article, the wildfire damaged areas are measured through using a kind of scenarios analysis for the climate and topographical factors of the Bayesian wildfire model. However, since there are very uncertain in timing of the wildfire incidents and the wildfire spread prediction, it would be worthwhile to assess the economic damages of the wildfire using extreme cases on lower and upper tails of probability distribution of each risk variable under the framework of the stress test.

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